Supplementary Material

MOKG-HGNN: A Multi-Scale Heterogeneous Graph Neural Network Integrating a Biomedical Knowledge Graph for Pan-Cancer Subtype Classification

Released with the code on Zenodo and cited from the main text as Tables S1–S3 and Fig. S1. iCluster labels are C1–C28; C24 (LAML) is absent.

Table S1: Per-class test performance (mean, s.d. for F1 in parentheses; 5 seeds). P and R are MOKG-HGNN's; F1 is given for MOKG-HGNN and for the MOGNN-TF baseline (GCN, 11 seeds) with their difference Δ; n is the test-set support. Doubles as the iCluster subtype legend.

			MOKG-HGNN			**MOGNN-TF**	
Code	**Tumour**	n	**P**	**R**	**F1**	**F1**	**ΔF1**
C1	STAD	10	0.663	0.560	0.576 (0.120)	0.709 (0.142)	−0.133
C2	BRCA	30	0.542	0.680	0.602 (0.021)	0.737 (0.077)	−0.135
C3	SARC	91	0.771	0.765	0.767 (0.047)	0.818 (0.039)	−0.051
C4	COAD	88	0.904	0.877	0.890 (0.011)	0.906 (0.027)	−0.016
C5	CNS/endocrine	53	0.768	0.785	0.775 (0.042)	0.803 (0.040)	−0.028
C6	OV	55	0.934	0.975	0.954 (0.024)	0.952 (0.026)	+0.002
C7	SARC	39	0.221	0.308	0.257 (0.031)	0.262 (0.068)	−0.005
C8	UCEC	91	0.870	0.822	0.845 (0.031)	0.890 (0.025)	−0.045
C9	ACC	27	0.893	0.948	0.917 (0.026)	0.904 (0.027)	+0.013
C10	LUSC	125	0.611	0.450	0.516 (0.053)	0.691 (0.029)	−0.174
C11	LGG	81	0.953	0.943	0.948 (0.010)	0.973 (0.015)	−0.025
C12	THCA	100	0.980	0.976	0.978 (0.020)	0.982 (0.008)	−0.004
C13	mixed (chr8 del)	42	0.332	0.467	0.387 (0.063)	0.458 (0.070)	−0.071
C14	LUAD	91	0.756	0.754	0.754 (0.040)	0.798 (0.024)	−0.044
C15	SKCM	61	0.813	0.866	0.838 (0.035)	0.852 (0.015)	−0.014
C16	PRAD	95	0.994	0.975	0.984 (0.016)	0.991 (0.006)	−0.007
C17	mixed	37	0.296	0.373	0.328 (0.049)	0.491 (0.080)	−0.163
C18	pan-GI (MSI)	57	0.545	0.761	0.634 (0.041)	0.769 (0.060)	−0.135
C19	BRCA	111	0.834	0.789	0.810 (0.028)	0.841 (0.032)	−0.031
C20	mixed (stromal/immune)	152	0.760	0.512	0.611 (0.041)	0.740 (0.031)	−0.129
C21	DLBC	6	0.663	0.900	0.752 (0.087)	0.799 (0.084)	−0.047
C22	SARC	37	0.717	0.735	0.723 (0.034)	0.775 (0.036)	−0.052
C23	GBM	10	0.665	0.860	0.745 (0.075)	0.696 (0.061)	+0.049
C25	pan-SCC (chr11 amp)	56	0.433	0.529	0.473 (0.082)	0.783 (0.054)	−0.310
C26	LIHC	65	0.911	0.997	0.952 (0.010)	0.971 (0.021)	−0.019
C27	CESC	114	0.769	0.719	0.743 (0.034)	0.809 (0.027)	−0.066
C28	KIRC	147	0.983	0.965	0.974 (0.010)	0.975 (0.011)	−0.001

Reproducibility

Pipeline (conda env `gnn`: Python 3.10, PyTorch 2.7 + PyG 2.7, CUDA 12.8):

```
setup_env.sh
prepare_data.sh                                  # download KG + TCGA, preprocess
make_graph.sh                                  # feature selection + build template
train_and_eval.sh                                # multi-seed training + evaluation
experiments.sh {per_class | explain}
run_feature_collapse.sh                                  # MODELS="mokghgnn mognntf"
```

Table S2: Feature collapse, full grid. Test macro-F1 and accuracy (mean, s.d.; 5 seeds) at each gene level, both models on the identical surviving molecular panel.

Model	Genes	F1 Macro	s.d.	Acc.	s.d.
MOKG-HGNN	700	0.7243	0.0106	0.7570	0.0147
	500	**0.7527**	0.0044	0.7877	0.0093
	300	0.7161	0.0062	0.7552	0.0041
	150	0.7013	0.0092	0.7395	0.0149
	100	0.7078	0.0092	0.7397	0.0055
	50	0.6949	0.0095	0.7348	0.0060
	20	0.6569	0.0123	0.6976	0.0104
MOGNN-TF	700	**0.7963**	0.0099	0.8285	0.0070
	500	0.7926	0.0212	0.8270	0.0115
	300	0.7728	0.0084	0.8061	0.0054
	150	0.7557	0.0076	0.7933	0.0066
	100	0.7340	0.0132	0.7804	0.0067
	50	0.7119	0.0200	0.7640	0.0128
	20	0.6386	0.0246	0.7252	0.0113

Table S3: Rank-1 distinctive mechanism per iCluster subtype (pathway and GO term), from GNNExplainer edge masks with degree normalisation. Names are truncated for width; full top-15 lists with distinctive scores are released with the code.

Subtype	Top pathway (rank-1)	Top GO term (rank-1)
C1/STAD	Transport of nucleosides and free purine an...	muscle contraction
C2/BRCA	Tie2 Signaling	transition metal ion binding
C3/SARC	Fibronectin matrix formation	metanephric nephron tubule for...
C4/COAD	Detoxification of Reactive Oxygen Species	methyl-CpG binding
C5	Voltage gated Potassium channels	regulation of blood pressure
C6/OV	NGF-independant TRKA activation	metanephric epithelium develo...
C7/SARC	Synthesis of 15-eicosatetraenoic acid deriv...	artery smooth muscle contract...
C8/UCEC	Estrogen biosynthesis	sex differentiation
C9/ACC	RAB geranylgeranylation	glial cell differentiation
C10/LUSC	Deactivation of the beta-catenin transacti...	transition metal ion binding
C11/LGG	DNA Damage/Telomere Stress Induced Senesce...	ionotropic glutamate receptor...
C12/THCA	Prolactin receptor signaling	leukotriene metabolic process
C13	Inositol transporters	liver regeneration
C14/LUAD	Defective SFTPA2 causes IPF	multivesicular body lumen
C15/SKCM	Methylation	vasodilation
C16/PRAD	Activated PKN1 stimulates transcription of...	regulation of systemic arteri...
C17	Latent infection – Other responses of Mtb...	prostate epithelial cord arbo...
C18	Transport of nucleosides and free purine an...	electron transport chain
C19/BRCA	Growth hormone receptor signaling	blood coagulation, fibrin clot...
C20	Chromatin modifying enzymes	intestinal epithelial cell dif...
C21/DLBC	Insulin processing	circadian rhythm
C22/SARC	Cellular Senescence	regulation of AMPA receptor ac...
C23/GBM	Insulin processing	photoreceptor cell maintenance
C25	Abacavir metabolism	carboxylic acid binding
C26/LIHC	Terminal pathway of complement	phospholipid efflux
C27/CESC	Activation of BAD and translocation to mito...	flavone metabolic process
C28/KIRC	Conjugation of phenylacetate with glutamine	fatty acid biosynthetic proce...

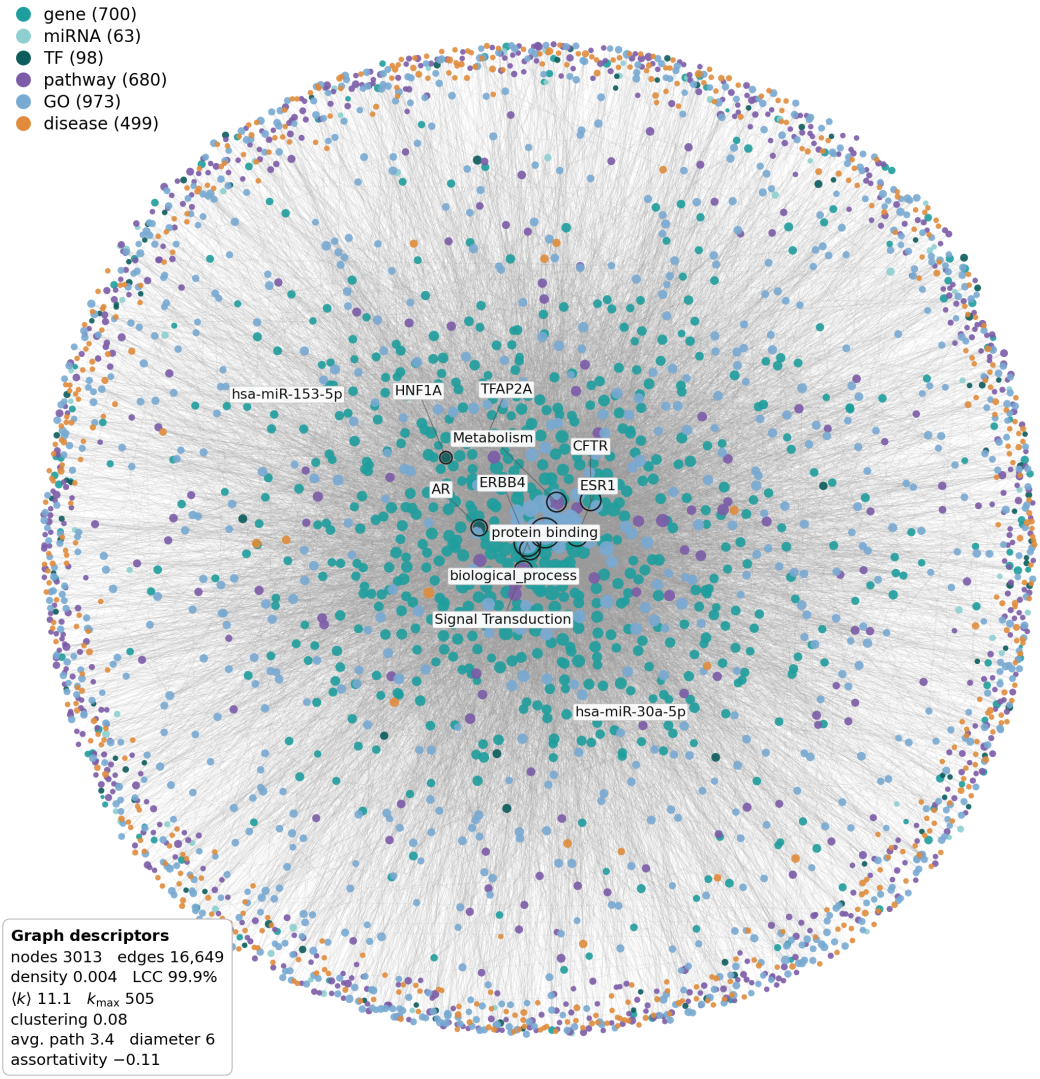


Figure S1: The multi-scale template after feature selection, the actual graph the model operates on. Nodes are the 3013 survivors across six biological scales (colour); node size is proportional to degree; only the highest-degree hub of each scale is labelled. Edges are the seven canonical relations tabulated in the main text; the layout is force-directed (Fruchterman–Reingold).